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ORIGINAL RESEARCH

Coronary Artery Calcium for Risk Stratification of Sudden Cardiac Death

The Coronary Artery Calcium Consortium

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ABSTRACT

OBJECTIVES Sudden cardiac death (SCD) is a common initial manifestation of coronary heart disease (CHD); however, SCD risk prediction remains elusive.

BACKGROUND Coronary artery calcium (CAC) is a marker of plaque burden. Whether CAC improves risk stratification for incident SCD beyond atherosclerotic cardiovascular disease (ASCVD) risk factors is unknown.

METHODS We studied 66,636 primary prevention patients from the CAC Consortium. Multivariable competing risks regression and C-statistics were used to assess the association between CAC and SCD, adjusting for demographics and traditional risk factors.

RESULTS The mean age was 54.4 years, 33% were women, 11% were of non-White ethnicity, and 55% had CAC >0. A total of 211 SCD events (0.3%) were observed during a median follow-up of 10.6 years, 91% occurring among those with baseline CAC >0. Compared with CAC = 0, there was a stepwise higher risk (P trend < 0.001) in SCD for CAC 100 to 399 (subdistribution hazard ratio [SHR]: 2.8; 95% CI: 1.6-5.0), CAC 400 to 999 (SHR: 4.0; 95% CI: 2.2-7.3), and CAC >1,000 (SHR: 4.9; 95% CI: 2.6-9.9). CAC provided incremental improvements in the C-statistic for the prediction of SCD among individuals with a 10-year risk <7.5% (ΔC-statistic = +0.046; P = 0.02) and 7.5% to 20% (ΔC-statistic = +0.069; P = 0.003), which were larger when compared with persons with a 10-year risk >20% (ΔC-statistic = +0.01; P = 0.54).

CONCLUSIONS Higher CAC burden strongly associates with incident SCD beyond traditional risk factors, particularly among primary prevention patients with low-intermediate risk. SCD risk stratification can be useful in the early stages of CHD through the measurement of CAC, identifying patients most likely to benefit from further downstream testing. (J Am Coll Cardiol Img 2022;■:■–■) © 2022 by the American College of Cardiology Foundation.

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ABBREVIATIONS AND ACRONYMS

ASCVD = atherosclerotic cardiovascular disease

CAC = coronary artery calcification

CHD = coronary heart disease

CT = computed tomography

HF = heart failure

ICD = International Classification of Diseases

SCD = sudden cardiac death

Sudden cardiac death (SCD) is characterized as an unexpected death within 1 hour of a change in clinical status among persons with or without preexisting atherosclerotic cardiovascular disease (ASCVD).[1] Overall, SCD contributes to between 180,000 and 450,000 deaths (7%-18%) annually, and accordingly is one of the leading causes of death in the United States.[2] Coronary heart disease (CHD) secondary to atherosclerosis is responsible for up to 80% of all SCD events[3] and fewer than 15% of cases occur among individuals with a known arrhythmia or heart failure (HF).[4] Despite significant improvements to the management of subclinical CHD in primary prevention patients, prediction of SCD remains elusive and is largely only initiated in secondary prevention populations after individuals have already experienced an initial ASCVD event.[5] Low-cost and readily available subclinical atherosclerosis imaging has emerged as an important modality for ASCVD risk assessment in primary prevention patients and may have the potential to refine SCD risk assessment much earlier in the causal disease pathway.

Coronary artery calcium (CAC) is the most commonly used marker of subclinical coronary atherosclerosis and is noninvasively measured on cardiac-gated noncontrast computed tomography (CT) via the Agatston method.[6] CAC is strongly associated with incident ASCVD events and is a recommended diagnostic imaging modality for adults >40 years of age when there is uncertainty in the 10-year predicted risk and initiation of primary prevention pharmacotherapy.[7,8] Although the role of CAC scoring is increasing for ASCVD risk stratification and the prediction of all-cause mortality, the utility of CAC for predicting SCD remains unknown.

The documented potential for discordance between measured traditional ASCVD risk factors and subclinical atherosclerosis[9,10] supports the need for the testing of novel strategies to predict SCD in the general population. It has been previously shown that the presence and extent of CHD is a predictor of SCD[11]; however, the use of a noninvasive subclinical atherosclerosis detection method for predicting SCD has not been pursued, especially in asymptomatic primary prevention patients. Here, we sought to assess the prospective association of CAC burden with SCD among persons without clinical ASCVD at baseline, and to determine how age and sex modify the relationship between CAC and future SCD.

METHODS

STUDY POPULATION. The CAC Consortium is a multicenter cohort study that includes 4 high-volume centers in the United States: Cedars-Sinai Medical Center (Los Angeles, California), PrevaHealth Wellness Diagnostic Center (Columbus, Ohio), Harbor-UCLA Medical Center (Torrance, California), and Minneapolis Heart Institute (Minneapolis, Minnesota). The multicenter retrospective cohort study aimed to assess the association between CAC and long-term, disease-specific mortality. The study design and methods have been previously described in detail elsewhere.[12] In brief, investigators included individuals >18 years of age who were primarily free of clinical ASCVD or cardiovascular symptoms at the time of CAC scanning. For the current study, we included all 66,636 participants in the CAC Consortium who were referred to undergo CAC scanning (1991-2010) because of the presence of underlying ASCVD risk factor(s) and uncertainty in long-term ASCVD risk. Written informed consent for participation in research was collected at each respective field center at the time of CAC scanning at baseline. Institutional Review Board approval for coordinating center actions was by the Johns Hopkins University School of Medicine

MEASUREMENT OF CAC. A standard protocol was used to quantify CAC using noncontrast, electrocardiogram-gated cardiac CT at all participating medical centers.[12] Electron-beam and multidetector CT were used for imaging protocols, and earlier assessments have demonstrated that there are no clinically significant differences in CAC quantification between the 2 different scanning methods. Calcium scores were computed using the Agatston method. Participants were categorized into 5 CAC score groups: CAC 0, CAC 1 to 99, CAC 100 to 399, CAC 400 to 999, and CAC >1,000.

OUTCOME ASCERTAINMENT. Mortality in the CAC Consortium was assessed by linking patient records with the Social Security Administration Death Master File using a previously validated algorithm. Similar to previous studies involving SCD, death certificates were obtained from the National Death Index and underlying cause of death was categorized into common causes of death using International Classification of Diseases (ICD)-9th and 10th Revision codes.[13-15] When compared with known deaths identified via the electronic medical record in a subsample of the CAC Consortium, there was up to 90% specificity and sensitivity for the identification of deaths.[12] Death rates in the CAC Consortium are similar when compared with US Census Bureau data and relevant population-based cohorts.[12]

Similar to previously conducted epidemiological studies,[13-15] SCD was defined as mortality caused by cardiac dysrhythmia (ICD-9: 427), cardiac arrest (ICD-10: I46), paroxysmal tachycardia (ICD-10: I47), and/or other cardiac arrhythmias (ICD-10: I49). For the current analysis, an SCD event was considered if a participant had 1 of the specific SCD ICD codes in the underlying cause position on the death certificate or if the SCD ICD code was in the first position of the death certificate when CHD was the underlying cause of death.

EVALUATION OF ASCVD RISK FACTORS. Assessment of ASCVD risk factors occurred contemporaneously with CAC testing. Diabetes and hypertension were defined by a previous clinical diagnosis or reported antihypertensive or glucose-lowering medication use. There was no information regarding the differentiation between type 1 versus type 2 diabetes. Dyslipidemia (low-density lipoprotein-cholesterol >160 mg/dL), hypertriglyceridemia (triglycerides >150 mg/dL), and low high-density lipoprotein-cholesterol (<40 mg/dL in men, <50 mg/dL in women) were defined by a previous clinical diagnosis or use of lipid-lowering therapy. Information on smoking and family history of CHD (first-degree relative with history of CHD at any age) were obtained through self-report data. The 10-year risk for ASCVD was calculated using the pooled-cohort equations,[16] including using the raw equations to extrapolate risk for those younger than 40 years. Multiple imputation was performed in the case of limited missing data by using nonmissing data on age, sex, race, ASCVD risk factors, and CAC data (28% of participants in the overall CAC Consortium cohort). Of those participants with missing data, most (>72%) were lacking information for only 1 demographic or risk factor variable.

As noted in the original CAC Consortium methodology paper, multiple imputation was performed on 28% of individuals who were missing data on ASCVD risk factors.[12] Missing observations were treated as missing at random, and the standard STATA command, "MI impute," was used in the imputation algorithm, which involved 10 imputations by default. The missing at random imputation method was used because the assumption was made that the probability of missingness of traditional ASCVD risk factors was not dependent on other missing values or other patterns of risk factor prevalence. Multiple imputation was performed controlling for observed nonmissing values of all nonmissing adjacent traditional ASCVD factors. As described previously,[12] mean and median 10-year ASCVD risk scores have had strong agreement when using nonimputed versus imputed CAC Consortium data.

STATISTICAL ANALYSIS. Study population characteristics were stratified according to CAC burden (CAC 0, CAC 1-99, CAC 100-399, CAC 400-999, and CAC >1,000). Given previous data suggesting that very high CAC (>1,000) in primary prevention patients confers an equivalent risk for an incident ASCVD event similar to that of stable secondary prevention patients,[17] an upper limit of CAC >1,000 was used in analyses. Continuous variables were presented as means and SDs, and percentages were used for categorical variables. Because of a non-normal distribution of CAC, the median was used to represent the central tendency of CAC scores. The Student's *t-test* and Wilcoxon signed-rank test were used to assess differences in normally and non-normally distributed continuous variables, respectively. Differences between categorical variables were evaluated through the chi-square test.

For associative analyses, we compared the number of SCD events across CAC burden categories (CAC 0, CAC 1-99, CAC 100-399, CAC 400-999, CAC >1,000). Although the original ASCVD risk groups consisted of 4 categories (0%-5%, 5%-7.5%, 7.5%-20%, >20%), we analyzed results across 3 guideline-based ASCVD risk groups (<7.5%, 7.5%-20%, >20%) to optimize statistical power in the setting of a limited number of SCD events. Overall, SCD events were expressed as absolute numbers and proportions for the overall study sample and according to CAC category. The total number of events was divided by person-years of follow-up to calculate SCD event rates (per 1,000-year follow-up). Kaplan-Meier survival curves were computed for SCD events for individuals with CAC 0, CAC 1 to 99, CAC 100 to 399, CAC 400 to 999, and CAC >1,000. Differences in survival among CAC burden groups were assessed through the log-rank test.

The associations between CAC burden and SCD were assessed through competing risks regression, with non-ASCVD mortality considered the competing risk. Similar to previously conducted analyses using SCD as an outcome,[18] we performed competing risk regression to account for non-SCD deaths. The base model was adjusted for age and sex. A subsequent model was additionally adjusted for diabetes, current cigarette smoking, hypertension, hyperlipidemia, and a family history of CHD. Statistical testing for a significant linear trend across increasing CAC burden categories was assessed by reporting the *P* value for the 5-level CAC term when evaluated as a continuous independent variable in a linear regression model.

The predictive value of CAC for future SCD events was evaluated through Harrel's C-statistic for all

TABLE 1 Baseline Characteristics of Study Participants in the CAC Consortium by CAC Score Categories

	Overall (N = 66,636)	CAC = 0 (n = 29,757)	CAC 1-99 (n = 20,534)	CAC 100-399 (n = 9,067)	CAC 400-999 (n=4,409)	CAC >1,000 (n = 2,869)
Age, y	54.4 ± 10.6	49.9 ± 9.21	54.9 ± 9.5	60.0 ± 9.5	63.0 ± 9.4	66.3 ± 9.7
Women	33.0	44.5	27.2	22.4	17.5	13.7
Race						
White	89.0	88.7	88.9	90.4	90.7	87.6
Asian	3.8	4.2	3.5	3.4	3.2	4.0
Black	2.3	2.3	2.4	2.1	1.5	3.1
Hispanic	3.1	3.3	3.3	2.6	2.8	3.4
Other	1.7	1.6	1.9	1.6	1.8	1.9
CAC Score, AU	164 ± 480	0 ± 0	28 ± 27	211 ± 85	638 ± 169	1,953 ± 1,152
CAC Score, AU	3 (0, 95)	0 (0, 0)	18 (5, 45)	193 (138, 275)	610 (489, 771)	1,572 (1,225, 2,256)
10-y ASCVD risk	7.4 ± 9.2	3.8 ± 4.7	7.3 ± 7.5	11.5 ± 10.3	15.1 ± 12.0	20.2 ± 14.9
10-y ASCVD risk category						
<7.5%	68.6	87.6	67.2	44.5	30.0	17.9
7.5%-20%	23.5	11.0	26.9	40.5	45.0	42.8
>20%	7.9	1.4	5.9	15.1	25.0	39.3
Hypertension	31.0	22.8	31.9	40.2	46.5	55.4
Dyslipidemia	54.4	48.0	56.6	60.9	65.1	67.3
Current smoker	9.6	8.9	9.7	11.0	11.0	10.2
Family history of CHD	46.1	45.6	46.4	46.2	46.5	48.5
Diabetes mellitus	6.8	3.9	6.6	9.6	12.8	19.5

Values are mean ± SD, median (Q1, Q3), or %.
AU = Agatston units; ASCVD = atherosclerotic cardiovascular disease; CAC = coronary artery calcification; CHD = coronary heart disease.

individuals and also according to ASCVD risk groups (low <7.5%, intermediate 7.5%-20%, high >20%). C-statistics were compared by testing the equality of receiver-operating characteristic areas using approaches derived by Wieand et al[19] for evaluating differences among nonparametric and parametric receiver-operating characteristic curves. Given the sex-specific distribution of CAC burden[20] and association between CAC with left ventricular function,[21] we performed 2 sensitivity analyses assessing: 1) the association between CAC and incident SCD separately in men versus women; and 2) the association between CAC and incident SCD after excluding individuals who had a diagnosis of HF-associated mortality.

Statistical analyses were conducted using STATA 17 (Stata Corp.). Statistical significance was defined as a *P* value <0.05 on a 2-tailed test.

RESULTS

The average age of participants was 54.4 years, 33% were women, 11% were non-White, and there were 36,879 persons (55.3%) with CAC >0. More than one-half of the participants (55.0%) had a 10-year ASCVD <5.0%, although hypertension (31.0%), dyslipidemia (54.4%), and a family history of CHD (46.1%) were particularly prevalent among study participants (Table 1). Except for current cigarette smoking and a family history of CHD, there was a consistently higher prevalence of ASCVD risk factors across increasing CAC burden categories.

There were 211 SCD deaths over a median follow-up of 10.6 years, and a total of 91% of all SCD events occurred in patients with CAC >0. The SCD rate was <0.50 per 1,000 person-years through 60 years of age, although individuals aged between 70 and 79 and those >80 years of age had SCD event rates of 1.49 and 5.04 per 1,000 person-years, respectively (Figure 1). The SCD event rate was consistently higher across increasing CAC burden categories in the overall sample and when stratifying by baseline ASCVD risk (Figures 2A to 2D). Compared with individuals with CAC = 0 (0.05 per 1,000 person-years), individuals with CAC 1 to 99 had a more than 2-fold higher SCD event rate (0.13 per 1,000 person-years), whereas this difference was considerably larger for those CAC 100 to 399 (0.48 per 1,000 person-years), CAC 400 to 999 (0.94 per 1,000 person-years), and CAC >1,000 (1.85 per 1,000 person-years).

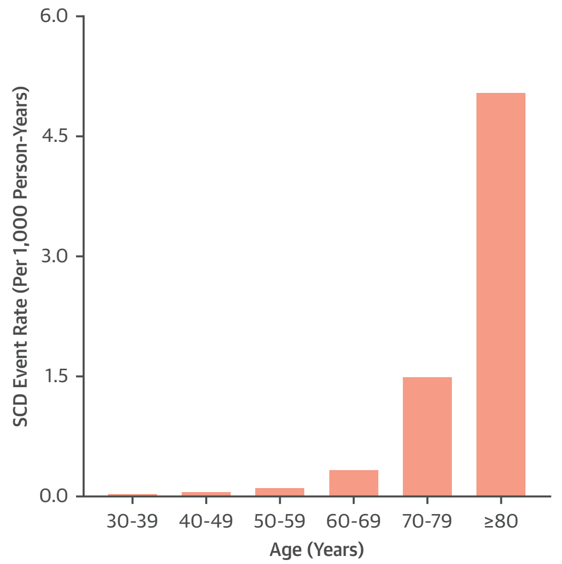
Kaplan-Meier curves showed divergence in SCD event rates as early as 2 years of follow-up, particularly for persons with CAC 100 to 399, CAC 400 to 999, and CAC >1,000 compared with those with CAC = 0 and CAC 1 to 99 (Figure 3A). Significant differences in SCD survival were consistently observed after stratifying by baseline ASCVD risk (Figures 3B to 3D). In fully adjusted models, there was a sequentially higher risk (*P* trend < 0.001) of SCD for increasing CAC burden (**Table 2**). Compared with individuals with CAC = 0, persons with CAC 1 to 399 were 2.8 times more likely to experience incident SCD (subdistribution hazard ratio [SHR]: 2.8; 95% CI: 1.6-5.0), whereas CAC 400 to 999 (HR: 4.0; 95% CI: 2.2-7.3) and CAC >1,000 conferred a 4-fold and 4.9-fold higher risk for SCD (HR: 4.9; 95% CI: 2.6-9.2), respectively (**Table 3 and Central Illustration**).

Although baseline 10-year ASCVD did not significantly modify the association between CAC and SCD (*P* interaction = 0.21), CAC scores 400 to 999 and >1,000 appeared nominally more strongly associated with SCD among persons with a risk <7.5% (SHR: 4.1; 95% CI: 1.3-13.4; SHR: 9.7; 95% CI: 2.9-31.7) and 7.5% to 20% (SHR: 3.9; 95% CI: 1.3-12.1; SHR: 7.4; 95% CI: 2.4-23.2) compared with those with a 10-year risk >20% (SHR: 3.6; 95% CI: 1.1-12.3; SHR: 3.6; 95% CI: 1.1-12.3). The difference in SCD risk between individuals with CAC 400 to 999 and those with CAC >1,000 was most notably appreciated among persons with low and intermediate 10-year ASCVD risk.

There was a similar SCD risk within a given CAC burden category for men versus women, and sex did not modify the association between CAC burden and SCD (Supplemental Table 1). Similar strengths of association between CAC burden and incident SCD were observed after excluding individuals who experienced HF-associated mortality (Supplemental Tables 2 and 3).

CAC scoring improved discrimination of SCD when added to traditional risk factors (**Table 4**). In addition to improving SCD prediction in the overall sample (C-statistic = 0.876 vs C-statistic = 0.858, Δ+0.018; *P* < 0.001), CAC scoring significantly improved SCD discrimination among persons spanning low, borderline, and intermediate 10-year ASCVD risks. CAC provided incremental improvements in the C-statistic for the prediction of SCD among individuals with a 10-year risk <7.5% (ΔC-statistic = +0.046; *P* = 0.02) and 7.5% to 20% (ΔC-statistic = +0.069; *P* = 0.003), which were considerably larger when compared with persons with a 10-year risk >20% (ΔC-statistic = +0.01; *P* = 0.54). Overall, the addition of CAC burden categories to traditional ASCVD risk factors provided a similar magnitude improvement in the C-statistic for incident SCD prediction in men and women (Supplemental Table 4). Significant improvements for incident SCD prediction were consistently noted for individuals with a 10-year ASCVD risk <7.5% and between 7.5% to 20% after excluding individuals who experienced HF-associated mortality (Supplemental Table 5).

FIGURE 1 Absolute SCD Event Rates



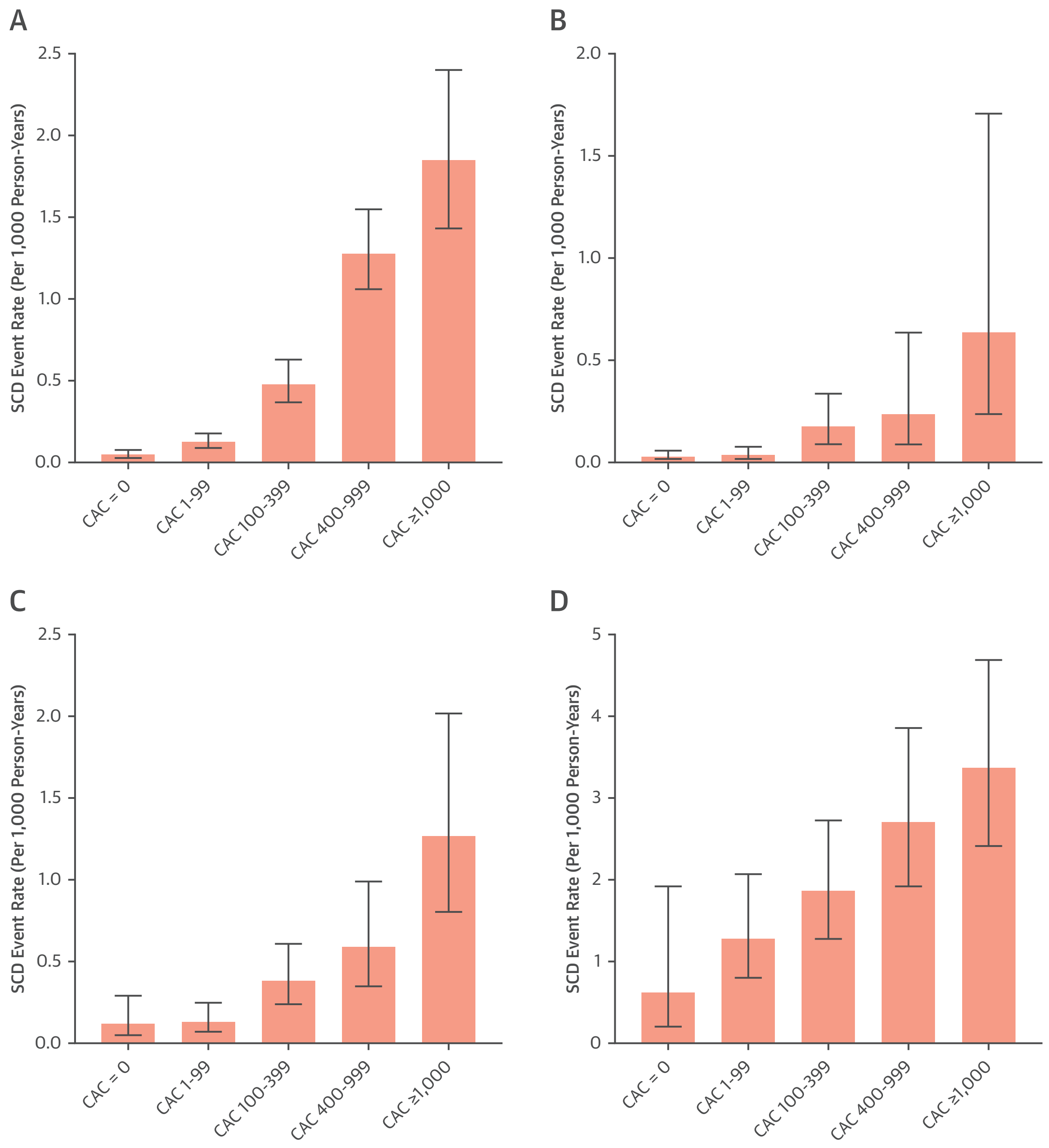
Absolute SCD event rates in the overall sample **(A)**, individuals with a 10-year ASCVD risk <7.5% **(B)**, 7.5%-20% **(C)**, and >20% **(D)**. The SCD event rate was consistently higher across increasing CAC burden categories in the overall sample and when stratifying by baseline ASCVD risk.
ASCVD = atherosclerotic cardiovascular disease; CAC = coronary artery calcification; SCD = sudden cardiac death.

DISCUSSION

Despite significant advancements in the management of subclinical CHD in primary prevention patients, the prediction of SCD remains elusive. Here, in a large sample of primarily asymptomatic patients undergoing CAC testing for risk stratification in primary prevention, we observed a stepwise increase in the SCD event rate across increasing CAC burden, such that middle-aged individuals with CAC scores 100 to 399, 400 to 999, and >1,000 had between a 2.8- to 4.9-fold higher risk for SCD compared with those with CAC = 0, independent of traditional risk factors. Contrary to the current clinical paradigm, SCD risk stratification may be most useful in the very early stages of CHD through the quantification of calcified atherosclerotic plaque burden, which may help to inform additional downstream genetic testing and imaging.

A large majority of SCD prevention efforts have focused on the burden of ventricular arrhythmias and/or SCD risk stratification after individuals have suffered an initial myocardial infarction.[5,22] These strategies will undoubtedly continue to remain important; however, the novelty of our study is that

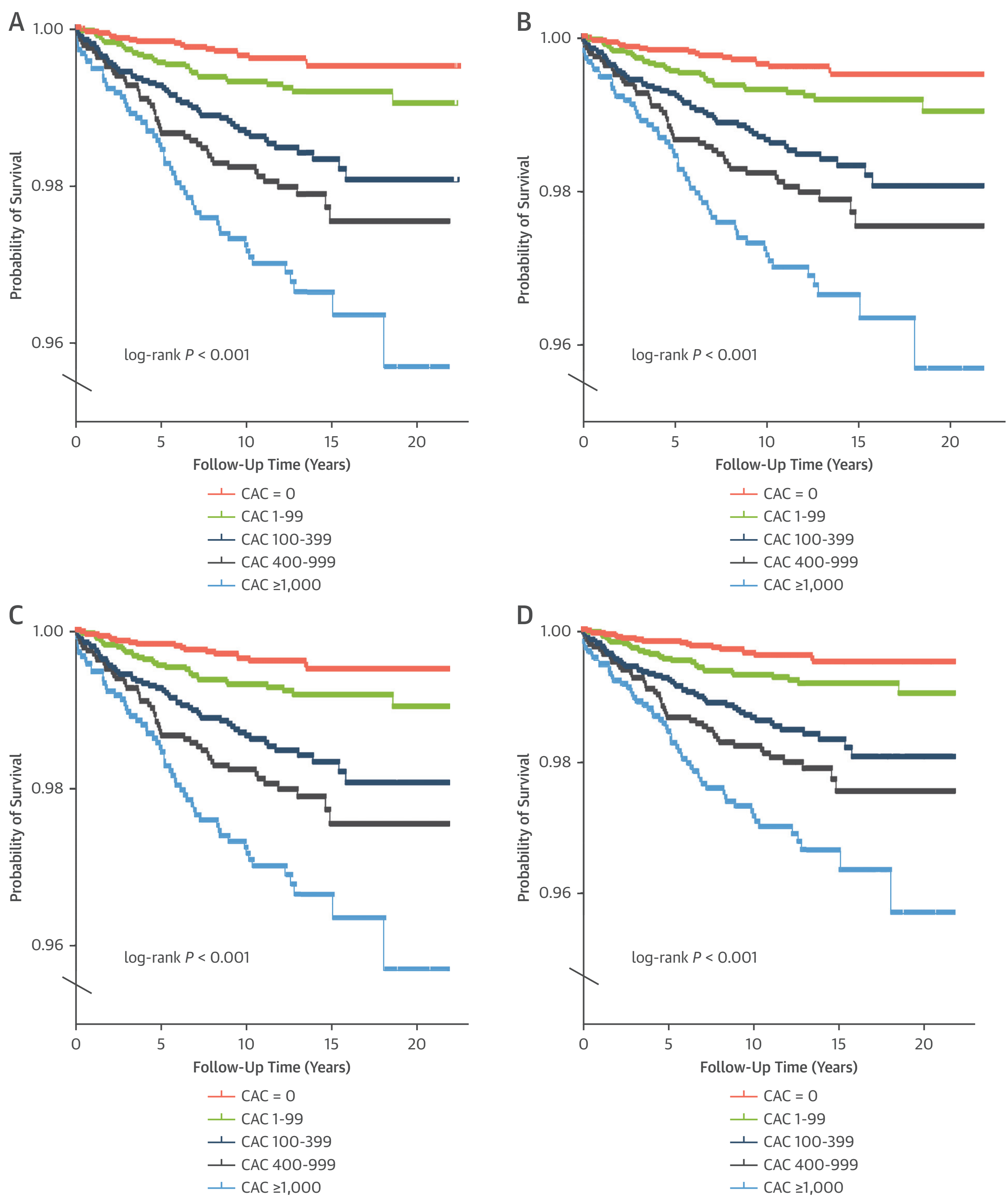
FIGURE 2 Absolute SCD Event Rates, Stratified by Age



SCD has a logarithmic association with age, strongly increasing in magnitude past 60 years old. Abbreviation as in Figure 1.

we demonstrate how an early approach focused on quantifying the spectrum of plaque burden among primary prevention patients can help to refine SCD risk stratification very early in the atherosclerotic process. Compared with electrophysiological predictors from prior studies, such as QT prolongation, QRS duration, and Cornell voltage criteria,[23-25] prevalent CAC >100 appeared more strongly associated with incident SCD in our primary prevention cohort. Approximately 4 of 5 SCD deaths are found to result from atherosclerotic CHD,[3] thus our study fulfills an unmet need to support identifying potential anatomical predictors for SCD, especially early in the disease process.

FIGURE 3 **Kaplan-Meier Plots For SCD Survival Probability**



Kaplan-Meier plots for SCD survival probability in the overall sample **(A)**, individuals with a 10-year ASCVD risk <7.5% **(B)**, 7.5%-20% **(C)**, and >20% **(D)**. Kaplan-Meier curves showed divergence in SCD event rates as early as 2 years of follow-up across CAC burden categories. Abbreviations as in Figure 1.

TABLE 2 Multivariable-Adjusted SHRs (95% CIs) for CAC Burden With SCD

CAC Score Group	Events (n = 211)	Unadjusted SHR (95% CI)	Unadjusted *P* Trend	Model 1 SHR (95% CI)[a]	Model 1 *P* Trend	Model 2 SHR (95% CI)[b]	Model 2 *P* Trend
CAC = 0	19	–	<0.001	–	<0.001	–	<0.001
CAC 1-99	33	2.5 (1.4-4.4)		1.4 (0.8-2.5)		1.3 (0.7-2.4)	
CAC 100-399	53	9.2 (5.5-15.6)		3.2 (1.8-5.6)		2.8 (1.6-5.0)	
CAC 400-999	49	17.8 (10.5-30.2)		4.7 (2.6-8.5)		4.0 (2.2-7.3)	
CAC >1,000	57	33.1 (19.7-55.7)		6.3 (3.4-11.8)		4.9 (2.6-9.2)	

[a]Adjusted for age and sex. [b]Adjusted for age, sex, current cigarette smoking, diabetes, hypertension, hyperlipidemia, and a family history of coronary heart disease.
CAC = coronary artery calcification; SCD = sudden cardiac death; SHR = subdistribution hazard ratio.

With respect to cardiac function, several previous studies have observed that left ventricular ejection fraction predicts incident SCD, and current guidelines recommend implantable cardiac defibrillator placement for: 1) persons with a prior myocardial infarction at least 40 days ago who have an ejection fraction <30%; and 2) persons with New York Heart Association-defined HF stage II and III who have an ejection fraction <35%.[5] Our findings are unique in that all participants were free of clinical ASCVD at baseline, and further confirm that there exists a continuum of risk between primary and secondary prevention, which may be most effectively assessed through CAC scoring.[17,26] A potential clinical question arising from of our findings is whether CAC plus ejection fraction may be a better predictor of SCD than either alone, and whether those who have CAC >400 and especially CAC >1,000 would potentially benefit from selective further downstream primary prevention screening, such as assessment of left ventricular function and/or specific treatment for heightened SCD risk. Although high CAC burden is not an established risk factor for abnormal left ventricular ejection fraction and/or vice-versa, individuals with very high CAC (≥1,000) for example, may potentially benefit from an assessment of ventricular function. Such considerations may be most important among patients in high-risk occupations or who are newly initiating an intensive exercise program. Alternatively, CAC scoring may also help to refine SCD risk after a newly abnormal echocardiogram in persons with no known ischemic etiology, as it is known that the long-term mortality benefit of implantable cardiac defibrillator placement in

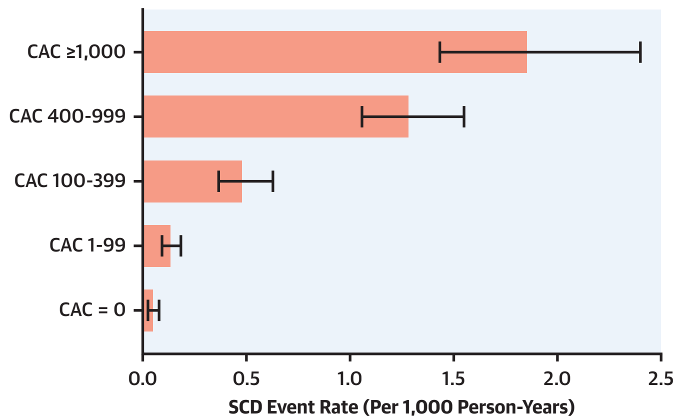
TABLE 3 Multivariable-Adjusted SHRs (95% CIs) for CAC Burden With SCD, Stratified by 10-Year ASCVD Risk

CAC Score Group	n	Unadjusted SHR (95% CI)	Model 1 SHR (95% CI)[a]	Model 2 HR (95% CI)[b]
ASCVD risk <7.5%	35			
CAC = 0		1	1	1
CAC 1-99		1.2 (0.5-3.1)	1.1 (0.4-2.9)	0.9 (0.3-2.5)
CAC 100-399		5.3 (2.2-12.6)	4.1 (1.6-10.5)	3.3 (1.3-8.4)
CAC 400-999		7.1 (2.3-22.3)	5.4 (1.6-18.6)	4.1 (1.3-13.4)
CAC >1,000		18.5 (5.9-58.0)	13.8 (4.1-46.9)	9.7 (2.9-31.7)
ASCVD risk ≥7.5%-20%	63			
CAC = 0		1	1	1
CAC 1-99		1.1 (0.4-3.2)	1.1 (0.3-3.3)	1.0 (0.3-3.2)
CAC 100-399		3.1 (1.1-8.4)	2.9 (1.0-8.5)	2.7 (0.9-7.8)
CAC 400-999		4.8 (1.7-13.2)	4.5 (1.5-13.7)	3.9 (1.3-12.1)
CAC >1,000		10.0 (3.7-26.9)	9.3 (3.0-28.4)	7.4 (2.4-23.2)
ASCVD risk ≥20%	113			
CAC = 0		1	1	1
CAC 1-99		2.0 (0.6-6.9)	2.1 (0.6-7.2)	2.0 (0.6-7.0)
CAC 100-399		2.9 (0.9-9.5)	2.7 (0.8-9.2)	2.7 (0.8-9.0)
CAC 400-999		4.1 (1.3-13.6)	3.8 (1.1-12.8)	3.6 (1.1-12.3)
CAC >1,000		4.8 (1.5-15.8)	4.0 (1.2-13.3)	3.6 (1.1-12.3)

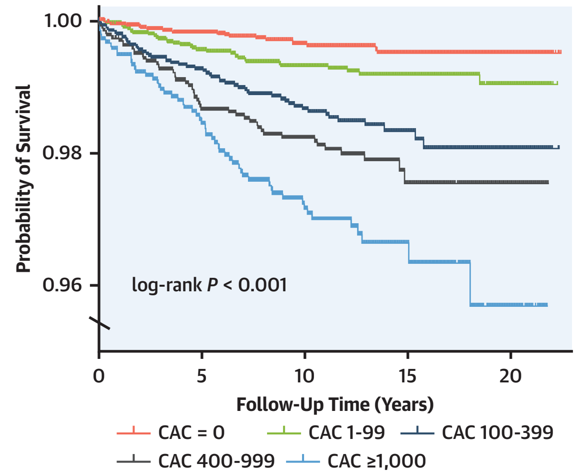
[a]Adjusted for age and sex. [b]Adjusted for age, sex, current cigarette smoking, diabetes, hypertension, hyperlipidemia, and a family history of coronary heart disease.
Abbreviations as in Table 1.

CENTRAL ILLUSTRATION Coronary Artery Calcium Burden and Incident Sudden Cardiac Death

Absolute SCD Event Rate According to CAC Burden



SCD Event Free Probability According to CAC Burden



Association Between CAC and Incident SCD Stratified by 10-Year ASCVD Risk

*when added to traditional risk factors	All		ASCVD Risk <7.5%		ASCVD Risk 7.5-20%		ASCVD Risk ≥20%	
	SHR (95% CI)	ΔC-Statistic*	SHR (95% CI)	ΔC-Statistic*	SHR (95% CI)	ΔC-Statistic*	SHR (95% CI)	ΔC-Statistic*
CAC = 0	Ref		Ref		Ref		Ref	
CAC 1-99	1.3 (0.7 – 2.4)		0.9 (0.3 – 2.5)		1.0 (0.3 – 3.2)		2.0 (0.6 – 7.0)	
CAC 100-399	2.8 (1.6 – 5.0)	+0.018	3.3 (1.3 – 8.4)	+0.046	2.7 (0.9 – 7.8)	+0.069	2.7 (0.8 – 9.0)	+0.008
CAC 400-999	4.0 (2.2 – 7.3)		4.1 (1.3 – 13.4)		3.9 (1.3 – 12.1)		3.6 (1.1 – 12.3)	
CAC ≥1,000	4.9 (2.6 – 9.2)		9.7 (2.9 – 31.7)		7.4 (2.4 – 23.2)		3.6 (1.1 – 12.3)	

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There is a stepwise increase in SCD event rates across increasing CAC burden, and SCD survival probability significantly differed as early as 2 years of follow-up across CAC burden categories. When added to traditional risk factors in the overall sample, CAC significantly improved the prediction of incident SCD.
ASCVD = atherosclerotic cardiovascular disease; CAC = coronary artery calcification; SCD = sudden cardiac death; SHR = subdistribution hazard ratio.

patients with nonischemic vs potentially ischemic cardiomyopathy appears to differ.[27]

Although the addition of CAC to traditional risk factors yielded the largest magnitude C-statistic improvements for SCD prediction among persons with low, borderline, and intermediate 10-year ASCVD risk compared with those with a high 10-year risk, the overall strength of association between CAC and SCD is perhaps the more important novel finding of our study. Whereas the C-statistic is a statistical tool to assess global discrimination, the stepwise increase in SCD risk across higher CAC burden suggests an underlying pathophysiological mechanism linking total calcified plaque burden and SCD, which could be mediated by tertiary variables such as left ventricular function, myocardial scar, and exercise-induced ischemia.

We identified a logarithmic association between age and incident SCD events, such that there was a large increase in SCD burden beginning at an age of 60 years. More than 80% of Westernized adults older than 65 years have CAC,[28] therefore leveraging the differences in CAC burden in this demographic group may be especially useful to predict and prevent incident SCD events. Given the age-specific pathobiology of SCD,[29] our results linking CAC testing to SCD may

TABLE 4 Discrimination Statistics of CAC for SCD Events Overall and Stratified by 10-Year ASCVD Risk

C-Statistic	Risk Factors[a]	CAC + Risk Factors	*P* Value
All	0.858	0.876	<0.001
ASCVD risk <7.5%	0.765	0.811	0.020
ASCVD risk 7.5-20%	0.708	0.777	0.003
ASCVD risk >20%	0.682	0.690	0.540

[a]Age, sex, current cigarette smoking, diabetes, hypertension, hyperlipidemia, and a family history of coronary heart disease.

Abbreviations as in Table 1.

thus be most generalizable in middle-aged and older persons compared with younger individuals. The absolute risk of SCD caused by coronary atherosclerosis, including progressive vascular calcification, increases in middle-aged and older persons versus younger individuals because the development of CAC occurs over several decades.[30] Thus, evaluation of SCD risk should expectedly look different as individuals age.

Although very low in magnitude, a total of 19 SCD events (0.06%) occurred among persons with CAC = 0, and the potential mechanisms underlying SCD in persons with CAC = 0 should be discussed. The prevalence of noncalcified atherosclerotic plaque burden has been reported to be as high as 11% in persons with CAC = 0,[31] which may increase the risk for potential plaque rupture or thrombosis leading to SCD. Alternatively, undiagnosed structural heart disease, coronary dissection or spasm, or other electrophysiologic predispositions could be potential mechanisms that explain the low residual SCD risk for individuals with CAC = 0.

The major strengths of this study include the measurement of CAC among more than 65,000 individuals who were enriched in ASCVD risk factors and a family history of CHD. As the cost of noncontrast CT decreases and the imaging modality becomes more readily available, further investigation regarding the association of CAC with downstream ASCVD outcomes will be increasingly important for refining risk assessment. Furthermore, we conducted one of the first associative analyses between CAC and SCD, and here we demonstrate that anatomic-based testing may be important for SCD risk stratification and prevention even among persons without overt clinical ASCVD who have an intermediate 10-year risk. Future research that assesses the association between CAC burden and SCD may extend on our findings by further phenotyping survival among sudden cardiac arrests and also using autopsy studies as a mechanism to assess validity in SCD ascertainment.

Our study should be considered in the setting of certain limitations. Given that the CAC Consortium is a primary prevention population, there was an overall low number of SCD events, and our study was underpowered for less-represented lower-risk groups including women. Similar to all studies ascertaining SCD as an outcome from death certificates and ICD codes, misclassification bias[32] and generalizability are potential limitations. For example, one of the main challenges that continues to affect all studies of SCD is that there is no universal standardized definition of SCD, including the duration of time from clinical onset of symptoms to cardiac arrest.[33] Such standardization will undoubtedly be necessary to improve precision in the prevention and risk assessment of SCD, particularly for identifying the subgroup of patients with low ASCVD risk that contribute to most SCD cases. Our findings relate specifically to the risk of SCD in screening populations and should not be misconstrued to represent the relationship of CAC scores to SCD among patients presenting with symptoms that are suggestive of coronary artery disease or in established secondary prevention. We used a validated method for SCD event ascertainment using National Death Index records to overcome this potential bias, and our results should be generalized only to middle-aged individuals who may have prevalent CAC and not used to guide SCD risk stratification in young adults. Furthermore, the CAC Consortium consists predominantly of participants of White ethnicity, and future studies involving CAC burden and SCD should incorporate greater ethnic diversity, specifically African American, Hispanic, and other non-White populations. Last, our statistical modeling did not account for multimodality testing and/or information on cardiac arrhythmias, as the CAC Consortium did not collect electrocardiogram, echocardiogram, stress testing data, or potential genetic channelopathies among patients.

CONCLUSIONS

A stepwise increase in SCD event rates occurred with increasing CAC burden. Among persons with low-to-intermediate 10-year ASCVD risk, CAC score categories from 100 through 1,000 independently conferred up to an approximate 5-fold higher risk for incident SCD compared with CAC = 0, and the addition of CAC improved incident SCD prediction when added to traditional risk factors. Overall, these results suggest that SCD risk stratification may be most useful in the very early stages of CHD through the quantification of calcified atherosclerotic plaque burden. Future studies that assess the role of CAC scoring as an initial modality to support the selective use of downstream testing, such as echocardiogram, and/or exercise treadmill testing with or without imaging or coronary CT angiography, may be useful to help further guide primary prevention strategies for SCD risk assessment.

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PERSPECTIVES

COMPETENCY IN MEDICAL KNOWLEDGE: CAC burden, particularly CAC >100, confers an increased risk for incident SCD independent of traditional ASCVD risk factors.

TRANSLATIONAL OUTLOOK: Future studies are required to assess the use of CAC scoring via noncontrast CT as an initial gatekeeper test guiding limited and selective use of additional noninvasive cardiac imaging/testing when advanced SCD risk stratification is required.

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KEY WORDS cardiovascular diseases, coronary artery calcium, multidetector computed tomography, sudden cardiac death

APPENDIX For supplemental tables, please see the online version of this paper.